EXPERIMENT – 9

❖ Computer-Aided Design and Optimization of a Microwave Impedance-Matching Network and Frequency-Selective Filter

Objectives of this Experiment:

1. To design an impedance-matching network for a specified complex microwave load.

2. To optimize the matching network for minimum reflection and maximum power transfer.

3. To design a microwave filter according to specified cutoff frequency, bandwidth and impedance requirements.

4. To evaluate return loss, insertion loss, passband, stopband and bandwidth using simulated frequency responses.

MATLAB CODE:

```
figure('Name','Microwave Impedance Matching',...
    'NumberTitle','off',...
    'Position',[100 100 1100 700]);


% Graph 1: Reflection Coefficient
subplot(2,2,1);
plot(f/1e9,abs(Gamma_before),'r--','LineWidth',2.5);
hold on;
plot(f/1e9,abs(Gamma_after),'b','LineWidth',2.5);
xline(10,'k:','10 GHz');
grid on;
grid minor;
xlim([8 12]);
ylim([0 0.4]);
xlabel('Frequency (GHz)');
ylabel('|\Gamma|');
```

```
title('Reflection Coefficient');
legend('Before Matching','After Matching','Location','best');
set(gca,'FontSize',10);

% Graph 2: Return Loss
subplot(2,2,2);
plot(f/1e9,-20*log10(abs(Gamma_before)),...
   'r--','LineWidth',2.5);
hold on;
plot(f/1e9,-20*log10(max(abs(Gamma_after),1e-6)),...
   'b','LineWidth',2.5);
xline(10,'k:','10 GHz');
grid on;
grid minor;
xlim([8 12]);
xlabel('Frequency (GHz)');
ylabel('Return Loss (dB)');
title('Return Loss');
legend('Before Matching','After Matching','Location','best');
set(gca,'FontSize',10);

% Graph 3: Input Impedance
subplot(2,2,3);
plot(f/1e9,real(Zin_before),'r--','LineWidth',2);
hold on;
plot(f/1e9,real(Zin_after),'b','LineWidth',2);
```

```
plot(f/1e9,imag(Zin_after),'k','LineWidth',2);
yline(50,'g:','50 Ohm');
xline(10,'k:');
grid on;
grid minor;
xlim([8 12]);
xlabel('Frequency (GHz)');
ylabel('Impedance (Ohm)');
title('Input Impedance');
legend('Before: Real','After: Real',...
   'After: Imaginary','50 Ohm Reference',...
   'Location','best');
set(gca,'FontSize',10);

% Graph 4: VSWR
subplot(2,2,4);
VSWR_before_f = (1+abs(Gamma_before)) ...
          ./(1-abs(Gamma_before));
VSWR_after_f = (1+abs(Gamma_after)) ...
          ./(1-abs(Gamma_after));

plot(f/1e9,VSWR_before_f,'r--','LineWidth',2.5);
hold on;
plot(f/1e9,VSWR_after_f,'b','LineWidth',2.5);
xline(10,'k:','10 GHz');
grid on;
```

```
grid minor;
xlim([8 12]);
xlabel('Frequency (GHz)');
ylabel('VSWR');
title('Voltage Standing Wave Ratio');
legend('Before Matching','After Matching','Location','best');
set(gca,'FontSize',10);

sgtitle('Microwave Impedance Matching Network: 8-12 GHz',...
    'FontSize',14,'FontWeight','bold');
```

OUTPUT

